Introduction to Machine Learning

Introduction to (Introduction to Machine Learning)

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IIT Kanpur

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First Things First

Administrative Stuff

- Please follow the rules of the CSE Department. Any destruction (for example writing stuff on benches) will result in the cancellation of registration.
- The course timings are 8:00-9:30 pm every day of the week. (including the weekends, sorry! I will not be availabe in the end of June and July)
- Every registered student should have an ID (badge), please keep it with you.
- The course homepage can be found at `http://sharbatc.github.io/teaching/`. All announcements/slides/other resources will be displayed on the homepage.

Personal Stuff

- My name is *Sharbatanu Chatterjee*. It's a difficult name to pronounce, I understand. It was made through cutting up and pasting different parts of my parents' names. It apparently has a Sanskrit etymology too, but enough linguistics for now.

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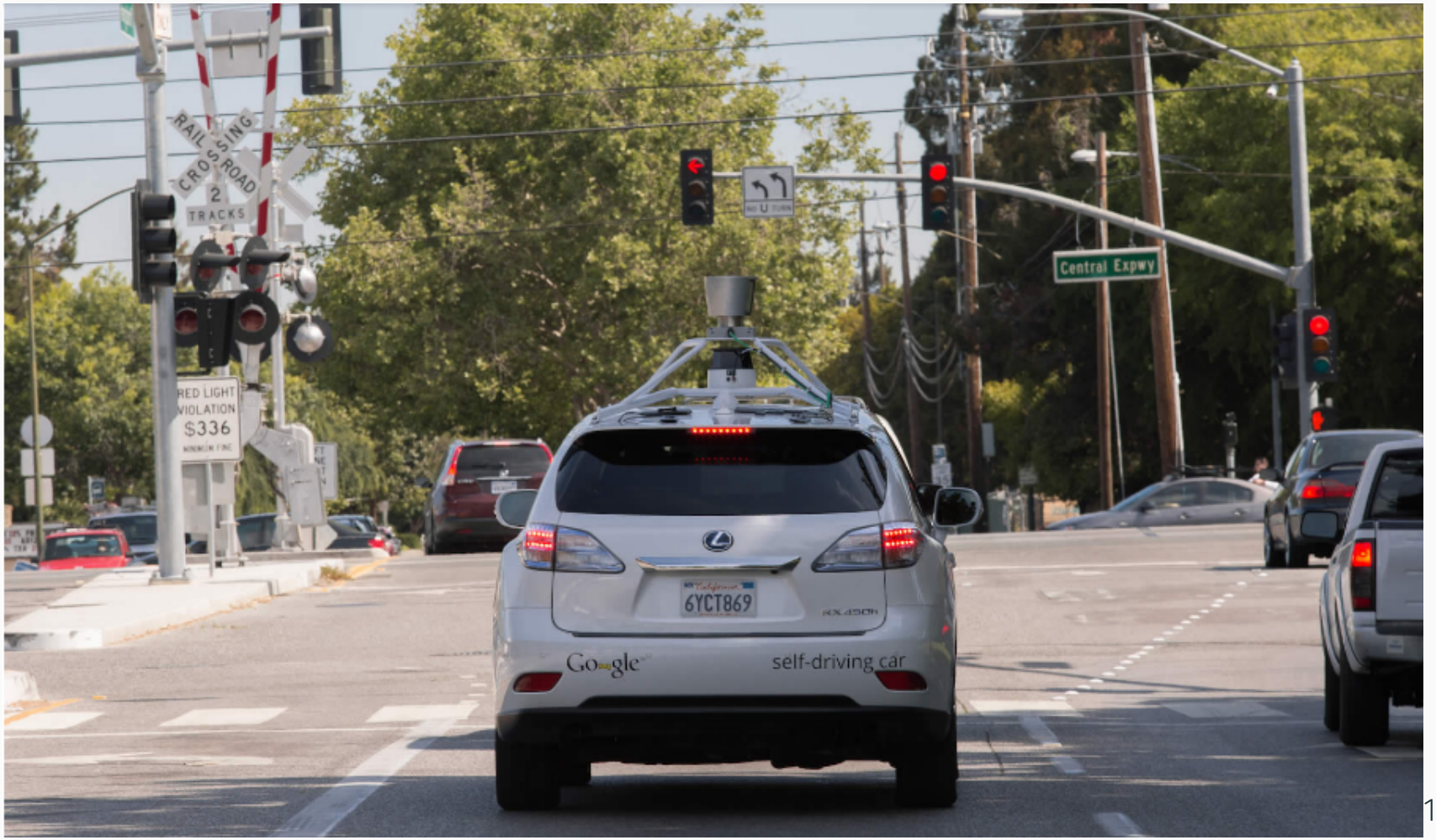
- The classes will be in English. However, please do tell me if you have problems understanding not only the concepts, but also my language. I will improve as I get feedback. I am also comfortable in Hindi, Bengali and a bit of Odia, so if you can put your questions better in these languages, please do not hesitate.

Introduction

Prof. Amitabha Mukerjee

Self Driving Car



[1]

[1] Image from `https://www.google.com/selfdrivingcar/where/`

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A self-driving car has the following:

- The Obstacle detection Unit
- The data stored in its computer
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The data consist of:

- Control instructions – it includes the driving strategy for autonomous mode
- Maps
- Sizes and shapes of various vehicles, how to differentiate between them and their behaviour on road
- The meaning of various road signs, traffic lights and the like User information – whom to let the car drive and whom to avoid

Self Driving Car

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For example, the obstacle detectors supplies the data of what is at what distance from the car and the sensors that are inside the car supply information like whether the driver is sleeping/awake or he has his hands on the steering or not, whether he is applying enough pressure on the brake/accelerator paddle and the like.

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After combining all these information, a driving strategy is made.

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- A field that studies algorithms that can learn from messy data and generalize. (Prof. Harish Karnick)
- Understand 'learning', biological and otherwise, and apply it to real-world problems. (Me)

History

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- Arthur Samuel (1952) made a checkers playing algorithm for the first time.
- Frank Rosenblatt (1959), working at Cornell, proposed an algorithm called the 'Perceptron', the first example of an artificial neuron. It was a part of the field of Cybernetics.
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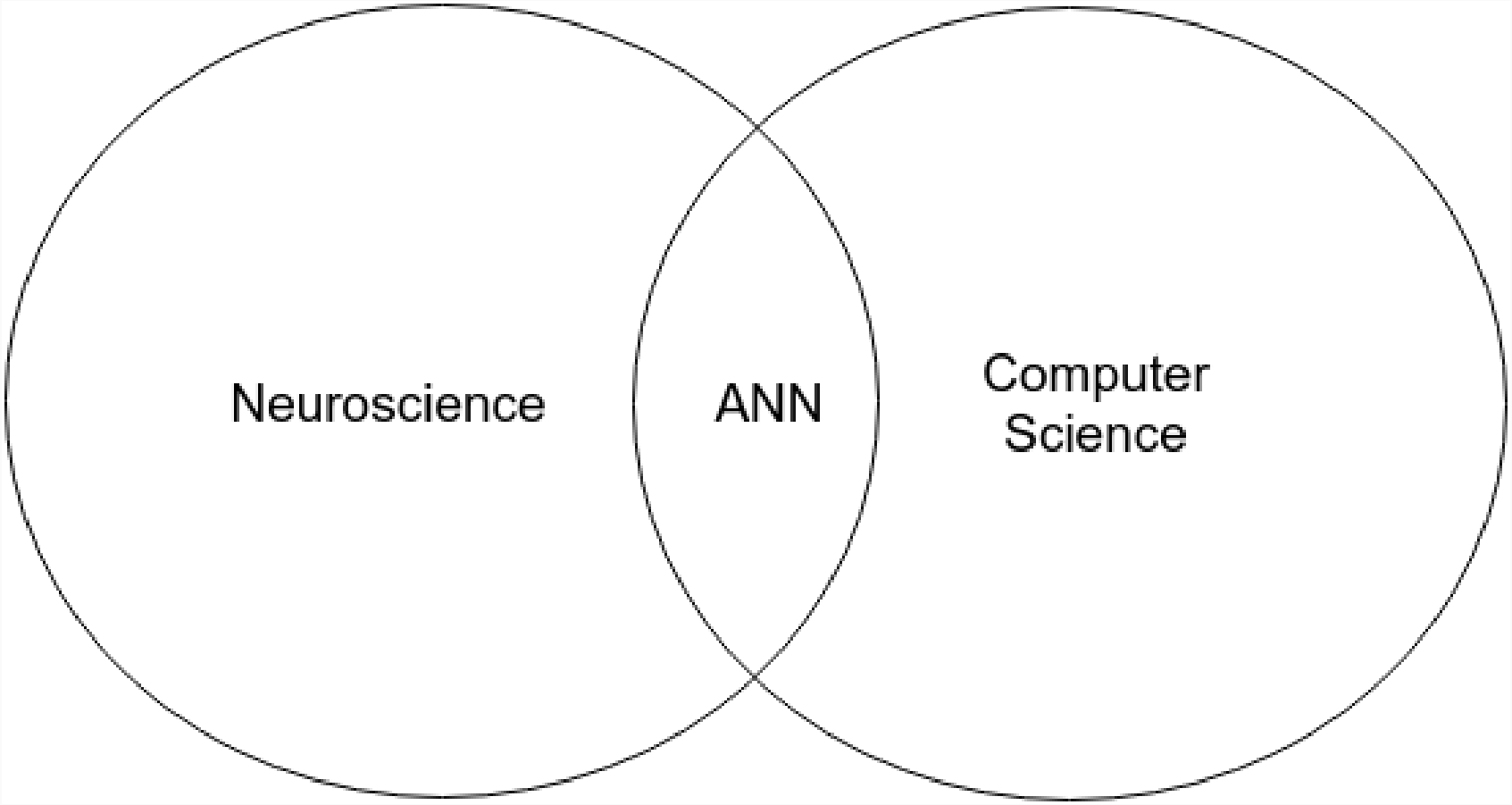
- Heavily inspired by the field of Brain and Neural sciences, which, by that time, was known to be the seat of intelligence in humans.
- However, an opposing emphasis emphasis on the logical, knowledge-based approach caused a rift between them and the neuroscientific people in AI.
- Probabilistic systems were plagued by theoretical and practical problems of data acquisition and representation.
- Minsky and Papert in *Perceptrons* (1969) controversially criticized the neural approach and brought ML to a halt (AI winter).

History



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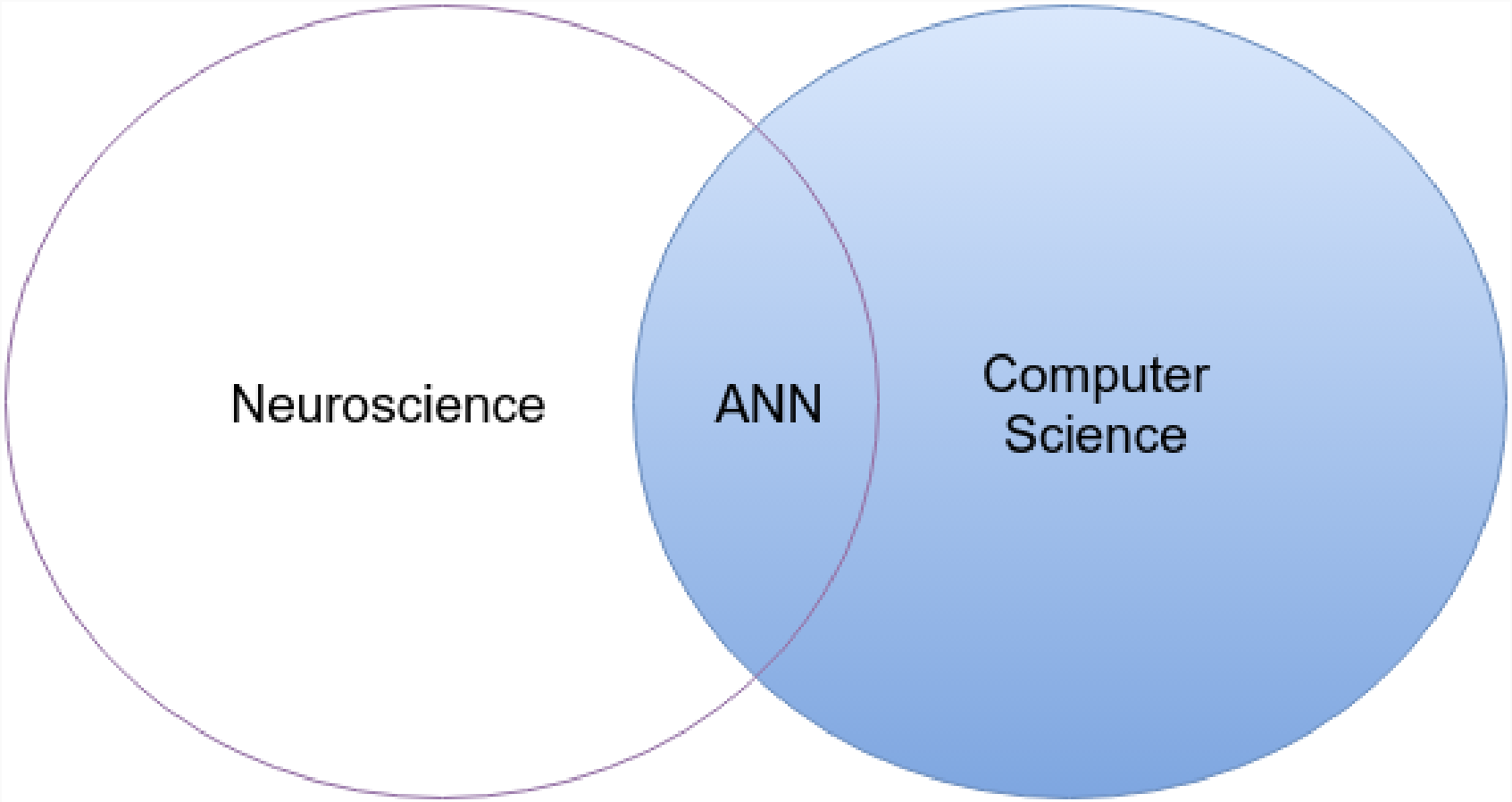

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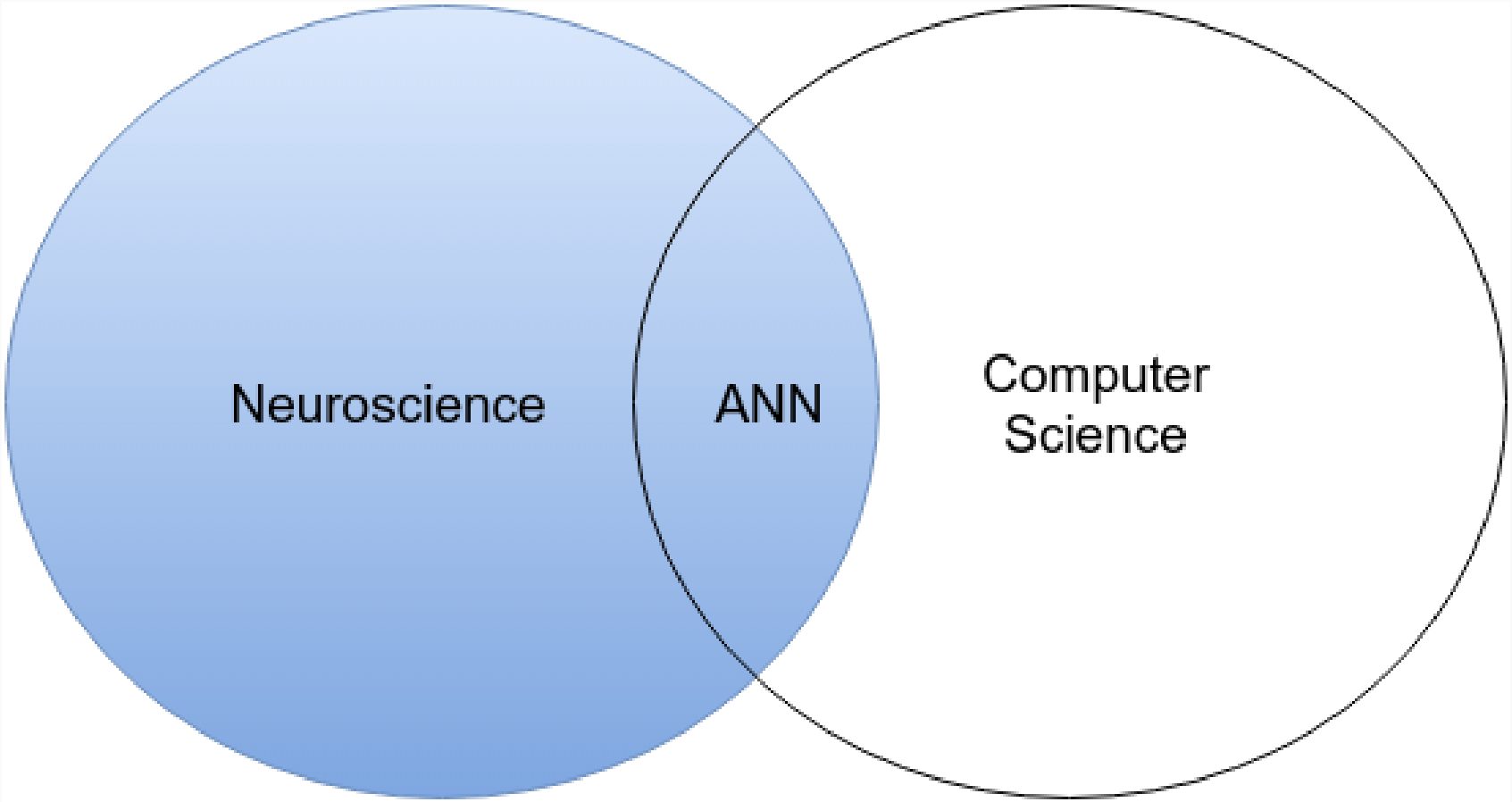

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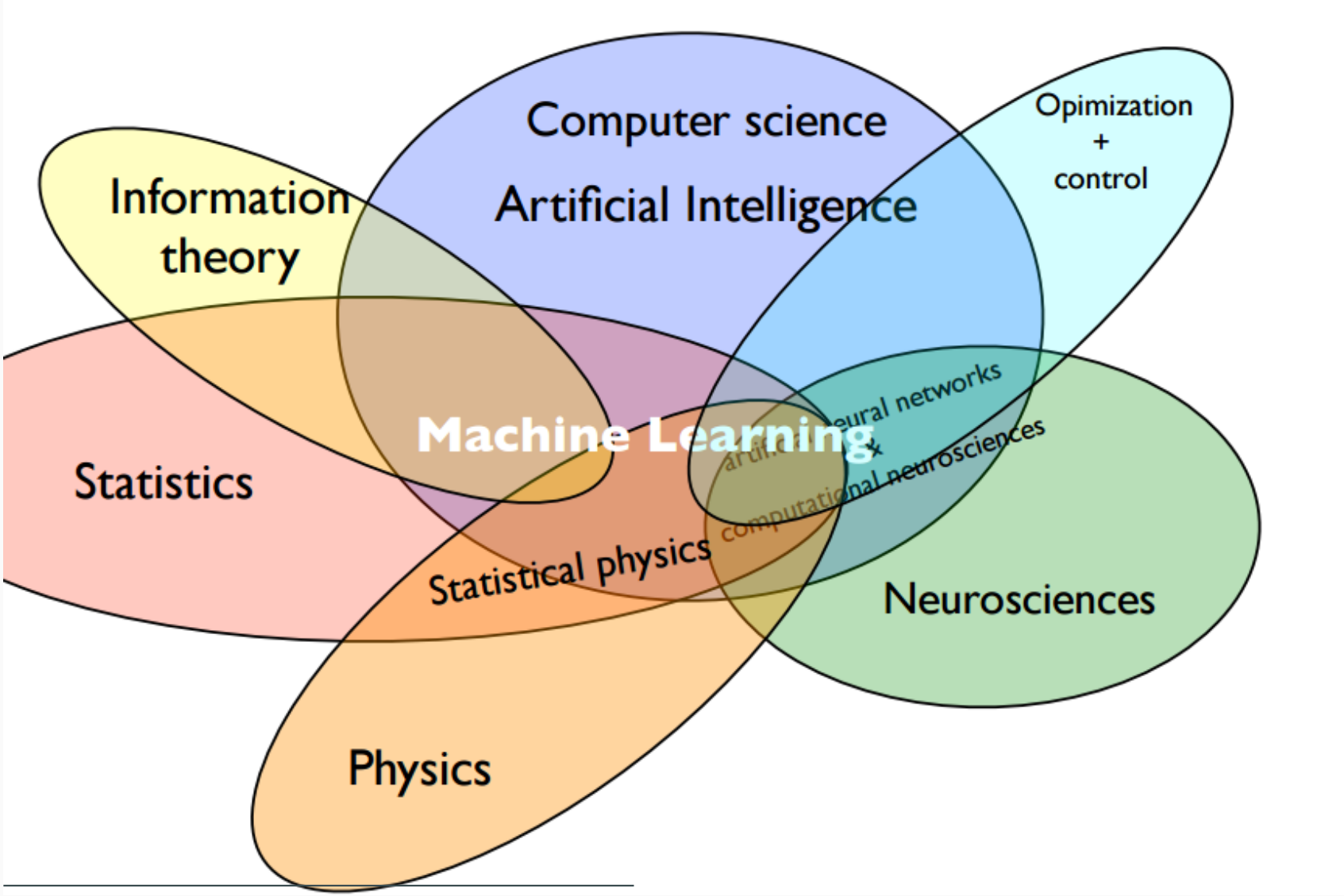

[2]

[2] Pascal Vincent, MILA, August 2015

Elements

Basic Element

What is the basic element of machine learning?

Data

Data is the currency that will feed your ML algorithms.

- Data is collected from various sources - nature, experiments, industry etc.
- They may be structured, unstructured, noisy, clustered etc.
- We would like to view data as a *list of examples*, ideally many of the same nature, often as a vector of numbers.
- If we have raw data, we will need to pre-process it to make it a vector of numbers, or whatever suits our needs.

Examples

One of the first famous successes of ML was in handwriting recognition by Yann LeCun et al.
(Check it out at `http://yann.lecun.com/exdb/mnist/`)

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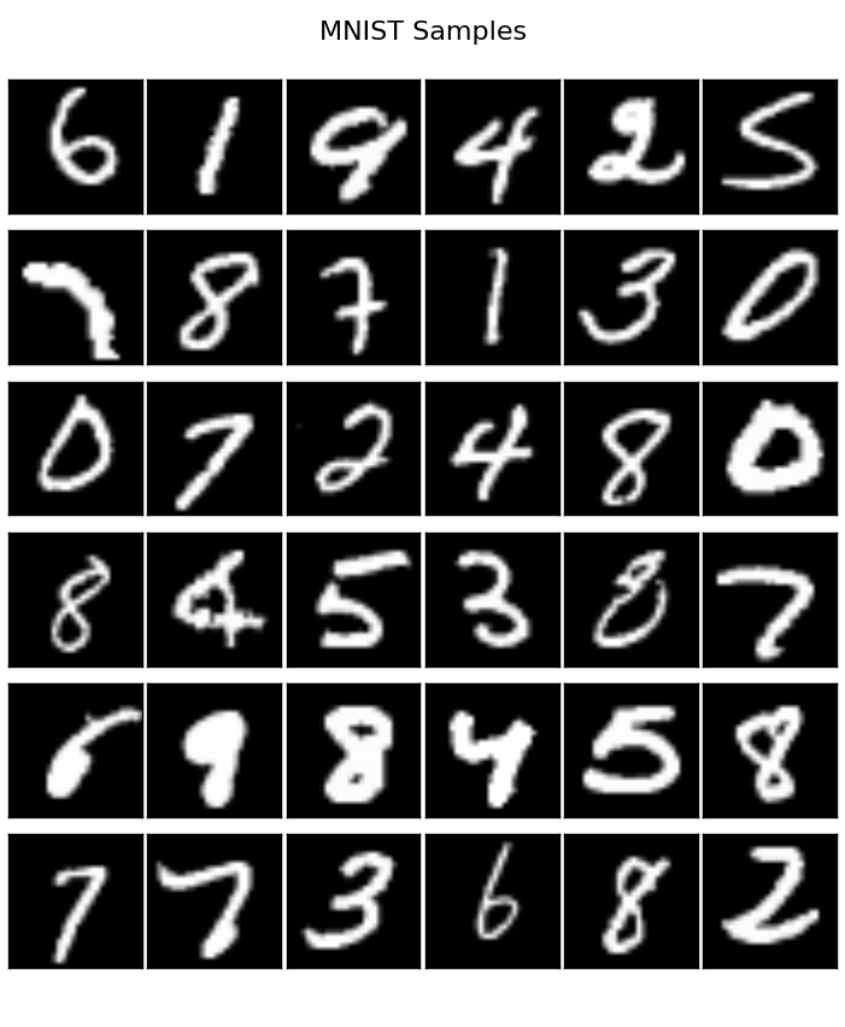


Examples

- So you see that we have a number of examples n, ranging to several millions
- Input dimensionality: d number of input features characterizing each example, can be 100 to 1000, even more.
- Target dimensionality, number of classes.

Examples : An example of visualization

Another way we can think of our problem is as a point in a d-dimensional vector space.

It is useful for some cases when we might want to cluster, visualize or work with distance measures.

Types of Problems

Types of Problems

The problems we deal with in Machine Learning can be classified into the following types:

- Supervised Learning
- Unsupervised Learning
- Reinforcement Learning

Supervised Learning

- Data can be divided into a set of independent and dependent variables.
- The goal is to learn a model that will predict the response variable.

Supervised Learning can be of several types.

- Classification (Binary, Multi-Class, Multi-Label). Examples are Spam Detection, Disease Diagnosis, OCR, Product Categories.
- Regression - Dependent variable here takes real values. Examples include predicting house prices, windspeeds, stock indices etc.

Unsupervised Learning

- This is meant to find some kind of structure in data, typically cluster some objects together.
- Needs the notion of similarity or dissimilarity metric
- Often used as a precursor to some other supervised learning problems

Let me mention in passing here about Reinforcement Learning, which is involved with deciding a number of sequential decisions to allow maximization of a 'reward function' in an environment influenced by your decisions.

Solving the Problems

Solving the Problems

The basic steps to solving any type of a machine learning problem:

- Learn a function f_θ that will act as a predictor.
- Validate the function such that it is constructed to perform reasonably well on the data we already have.
- Use the trusted f_θ now for new data, mot already seen

Caution

However, be careful that we do not 'validate' it too well, trusting our training data too much will lead to something called *generalization error*.

Solving the Problems

We may more formally say that the involved steps are, implicitly or explicitly:

- The choice of a function family
- A way to evaluate it (Loss Function)
- A way to find the best evaluation (Optimization)

Our task will be using this framework to minimize the Generalization Error, using whatever algorithm we deem fit for the situation

Solving the Problems

Minimizing the Generalization Error (also called Risk) is difficult. Because:

- All our data is/can be assumed to be taken from an unknown distribution, $p(x, y)$
- The Expected Risk is how poorly we will do on average on the infinity of future examples from that unknown distribution.

$$R_f = \mathbb{E}_{p(x,y)}[L(f(x), y)]$$

- The distribution is, well, unknown. Sorry. Hence we use Emperical Risk, defined as :

$$\hat{R}_{f,D} = \frac{1}{|D|} \sum_{(x,y)\in D} L(f(x), y)$$

Solving the Problems

So simply, find a training set, and find the function that minimizes the Empirical Risk over the said set.

$$\hat{f}(D_{train}) = \underset{f \in F}{\operatorname{argmin}}\, \hat{R}(f, D_{train})$$

Is that enough? Have we estimated the Generalization Error (Expected Risk) yet?

Solving the Problem

To get an unbiased estimate, we will need a separate test set.

$$R(\hat{f}(D_{train})) \approx \hat{R}(\hat{f}(D_{train}), D_{test})$$

Solving the Problem

All this while, we have talked of minimizing the error, given the function family we are supposed to be talking about.
How do we find the function family F?

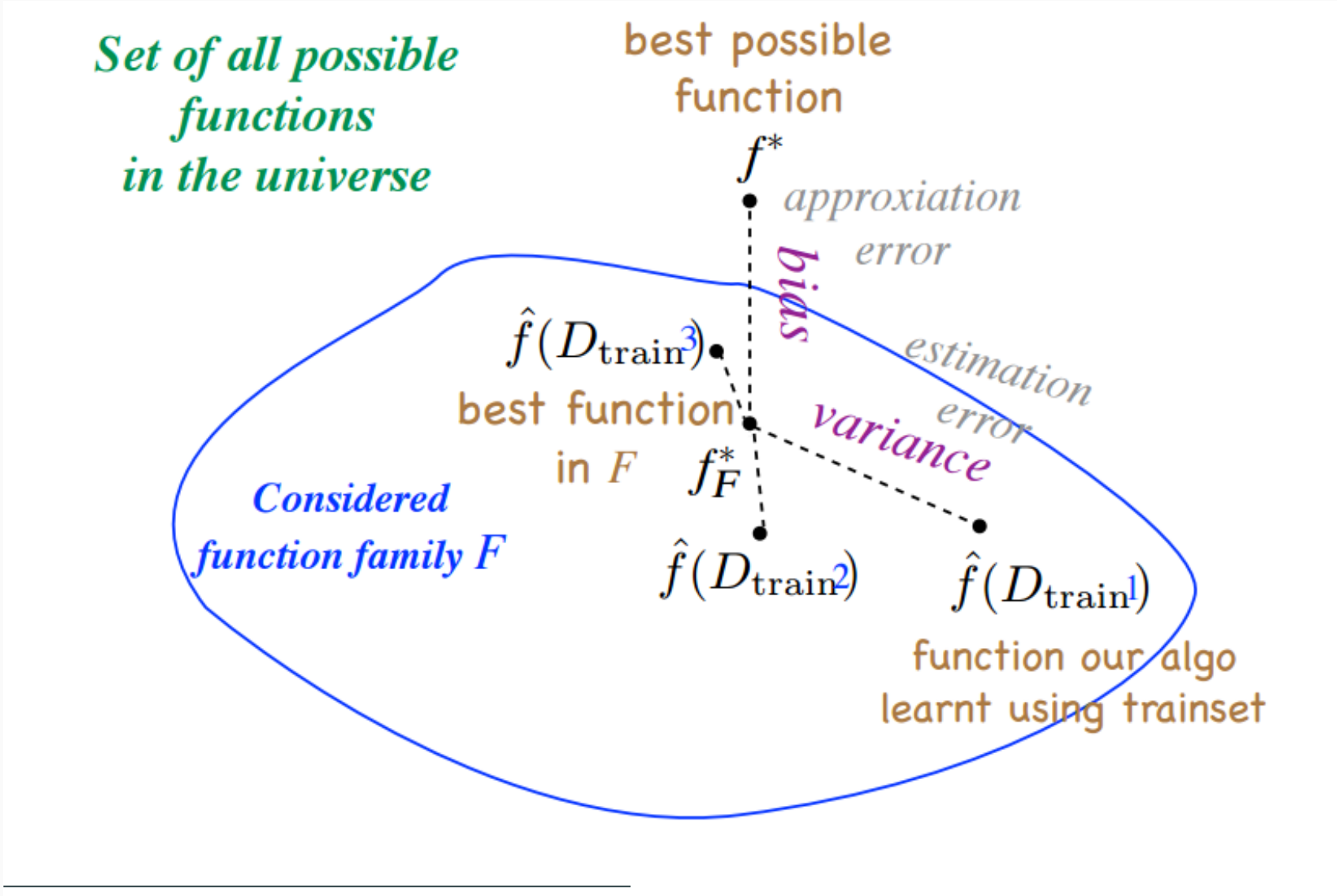
- Your ML algorithm is, in retrospect, the function family F that you choose.
- How expressive, or rich or powerful your ML algorithm is, depends on something that is informally called the capacity of the learning algorithm.
- Learning theory gives us several measures of capacity like the V-C dimension.
- One can estimate the capacity or complexity by finding the number of adaptable parameters that are contained in the model.

Solving the Problem

Why is capacity important?

- Capacity tells us how we selective our family is, with hyper-parameters and all.
- Tuning the capacity helps us choose sensible models and prevent over fitting and under-fitting.
- To choose hyper-parameters (the huge abundance of these are what determines the capacity), we will use a simple method - validation.

Solving the Problem



[3]

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FINALLY Solving the Problem

- For each model in our F, A and for each considered hyperparameter λ, do the following:
- Train model on D_{train} : $\hat{f}_{A_\lambda} = A_\lambda(D_{train})$
- Evaluate according to some evaluator, use the validation set for this. $e_{A_\lambda} = \hat{R}(f_{A_\lambda}, D_{valid})$
- Optimize to find the best A^*, λ^*
- Find f^*
- Find unbiased estimate of generalization performance on f^* as $\hat{R}(f^*, D_{test})$

Conclusion

Summary

So, we learnt:

- You will be given a dataset L to work with.
- You will need to come up with a model based on this to understand similar worldviews and datasets.
  At the end of this session, I would like you to internalize the concept of learning with data, and perhaps pick up and solve a problem from `https://www.kaggle.com/`

Questions?

Mathematical Preliminaries

- Vectors and Vector Spaces
- Matrices
- Arithmetic Operations
- Transposes
- Inverses
- Eigen Values
- Probability